In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load CSV (use your exact file name)
df = pd.read_csv("SuperMarket Analysis.csv")

df.head()
# Basic info
print(df.info())
print(df.describe())
print(df.isna().sum())

# Convert Date column to datetime
df["Date"] = pd.to_datetime(df["Date"])
# Total and average sales
total_sales = df["Sales"].sum()
avg_sales   = df["Sales"].mean()

sales_by_product = df.groupby("Product line")["Sales"].sum()
sales_by_gender  = df.groupby("Gender")["Sales"].sum()
sales_by_payment = df.groupby("Payment")["Sales"].sum()

print("Total sales:", total_sales)
print("Average sale:", avg_sales)
print("\nSales by product line:\n", sales_by_product)
print("\nSales by gender:\n", sales_by_gender)
print("\nSales by payment:\n", sales_by_payment)

# Total and average sales
total_sales = df["Sales"].sum()
avg_sales   = df["Sales"].mean()

sales_by_product = df.groupby("Product line")["Sales"].sum()
sales_by_gender  = df.groupby("Gender")["Sales"].sum()
sales_by_payment = df.groupby("Payment")["Sales"].sum()

print("Total sales:", total_sales)
print("Average sale:", avg_sales)
print("\nSales by product line:\n", sales_by_product)
print("\nSales by gender:\n", sales_by_gender)
print("\nSales by payment:\n", sales_by_payment)
plt.figure(figsize=(8,4))
sales_by_product.plot(kind="bar")
plt.title("Total Sales by Product Line")
plt.ylabel("Sales")
plt.xlabel("Product line")
plt.tight_layout()
plt.show()
plt.figure(figsize=(6,4))
sales_by_payment.plot(kind="bar")
plt.title("Sales by Payment Method")
plt.ylabel("Sales")
plt.xlabel("Payment")
```

```
plt.tight_layout()
plt.show()
plt.figure(figsize=(6,4))
sns.boxplot(x="Gender", y="Sales", data=df)
plt.title("Sales by Gender")
plt.tight_layout()
plt.show()
plt.figure(figsize=(8,6))
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 17 columns):
 #   Column                   Non-Null Count  Dtype  
---  ------                   --------------  -----  
 0   Invoice ID               1000 non-null   object 
 1   Branch                   1000 non-null   object 
 2   City                     1000 non-null   object 
 3   Customer type            1000 non-null   object 
 4   Gender                   1000 non-null   object 
 5   Product line             1000 non-null   object 
 6   Unit price               1000 non-null   float64
 7   Quantity                 1000 non-null   int64  
 8   Tax 5%                   1000 non-null   float64
 9   Sales                    1000 non-null   float64
 10  Date                     1000 non-null   object 
 11  Time                     1000 non-null   object 
 12  Payment                  1000 non-null   object 
 13  cogs                     1000 non-null   float64
 14  gross margin percentage  1000 non-null   float64
 15  gross income             1000 non-null   float64
 16  Rating                   1000 non-null   float64
dtypes: float64(7), int64(1), object(9)
memory usage: 132.9+ KB
None
        Unit price     Quantity      Tax 5%        Sales          cog
s  \
count  1000.000000  1000.000000  1000.000000  1000.000000  1000.0000
0
mean     55.672130     5.510000    15.379369   322.966749   307.5873
8
std      26.494628     2.923431    11.708825   245.885335   234.1765
1
min      10.080000     1.000000     0.508500    10.678500    10.1700
0
25%      32.875000     3.000000     5.924875   124.422375   118.4975
0
50%      55.230000     5.000000    12.088000   253.848000   241.7600
0
75%      77.935000     8.000000    22.445250   471.350250   448.9050
0
max      99.960000    10.000000    49.650000  1042.650000   993.0000
```

```
0

       gross margin percentage  gross income      Rating
count              1000.000000   1000.000000  1000.00000
mean                  4.761905     15.379369     6.97270
std                   0.000000     11.708825     1.71858
min                   4.761905      0.508500     4.00000
25%                   4.761905      5.924875     5.50000
50%                   4.761905     12.088000     7.00000
75%                   4.761905     22.445250     8.50000
max                   4.761905     49.650000    10.00000
Invoice ID                 0
Branch                     0
City                       0
Customer type              0
Gender                     0
Product line               0
Unit price                 0
Quantity                   0
Tax 5%                     0
Sales                      0
Date                       0
Time                       0
Payment                    0
cogs                       0
gross margin percentage    0
gross income               0
Rating                     0
dtype: int64
Total sales: 322966.749
Average sale: 322.966749

Sales by product line:
 Product line
Electronic accessories    54337.5315
Fashion accessories       54305.8950
Food and beverages        56144.8440
Health and beauty         49193.7390
Home and lifestyle        53861.9130
Sports and travel         55122.8265
Name: Sales, dtype: float64

Sales by gender:
 Gender
Female    194671.8375
Male      128294.9115
Name: Sales, dtype: float64

Sales by payment:
 Payment
Cash           112206.570
Credit card    100767.072
Ewallet        109993.107
Name: Sales, dtype: float64
Total sales: 322966.749
Average sale: 322.966749
```

```
Sales by product line:
 Product line
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Sales by payment:
 Payment
Cash           112206.570
Credit card    100767.072
Ewallet        109993.107
Name: Sales, dtype: float64
```
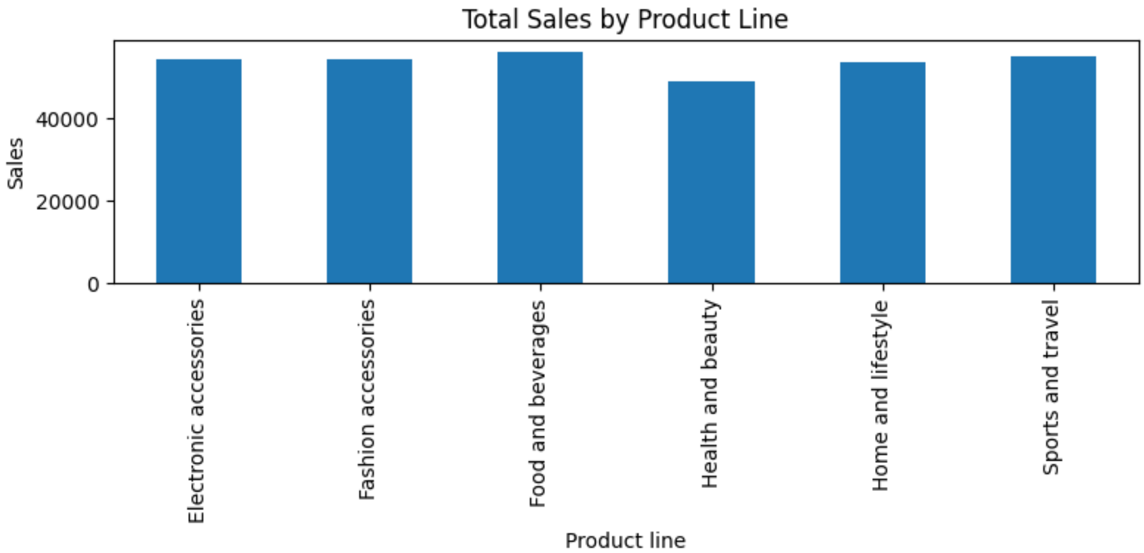
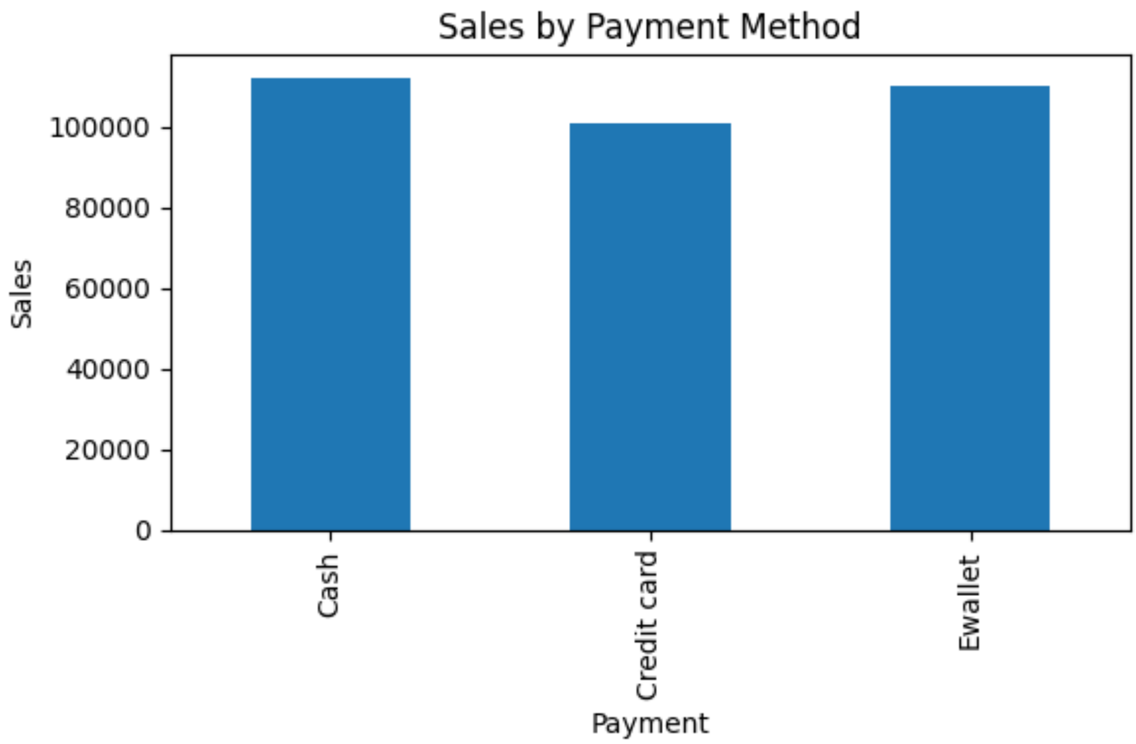
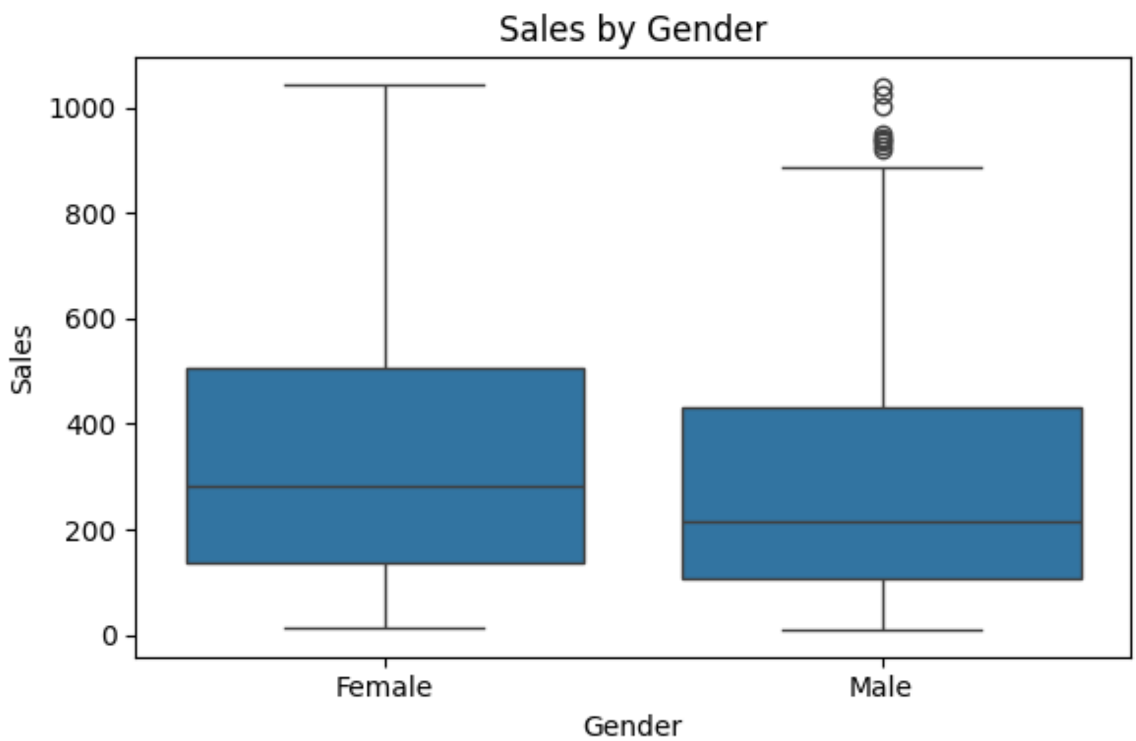
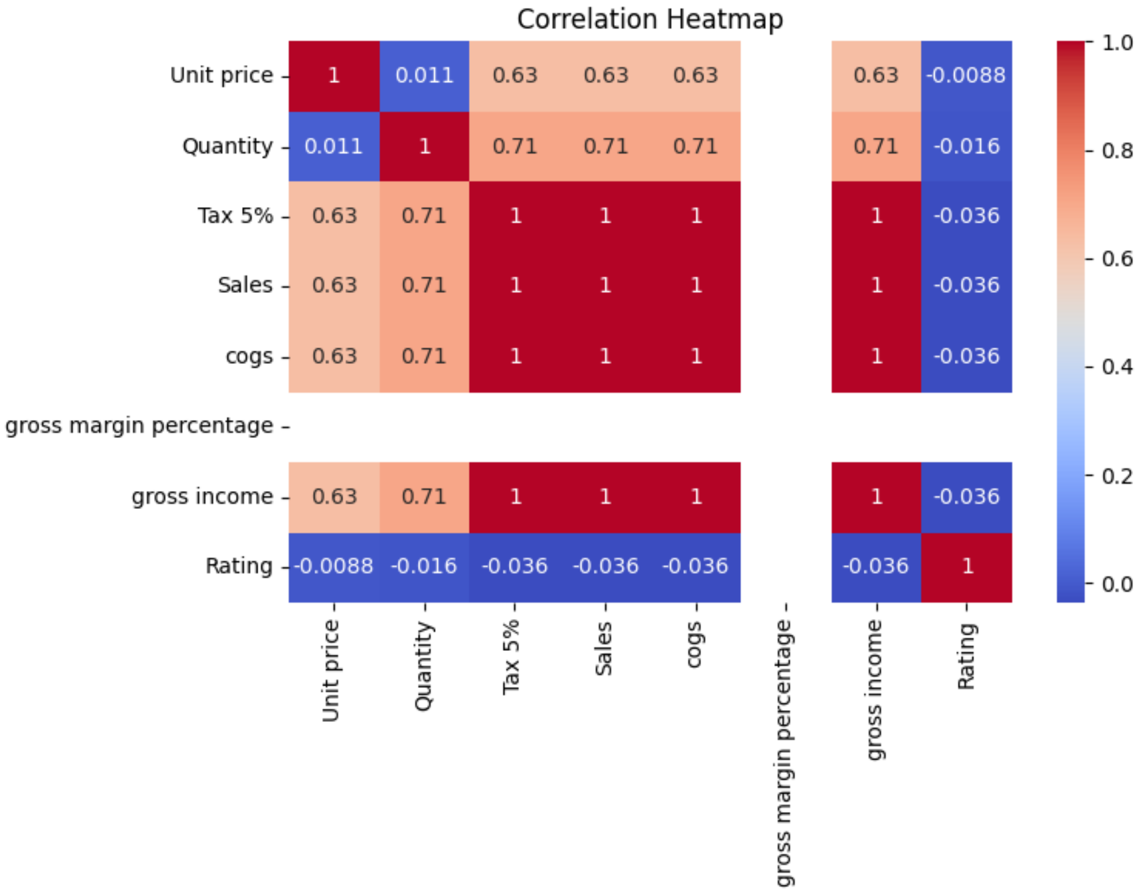




Sales by Payment Method
Sales
100000
80000
60000
40000
20000
0
Cash
Credit card
Ewallet
Payment


Sales by Gender
Sales
1000
800
600
400
200
0
Female
Male
Gender


Correlation Heatmap
Unit price
Quantity
Tax 5%
Sales
cogs
gross margin percentage
gross income
Rating
1
0.011
0.63
0.71
-0.0088
-0.016
-0.036
1.0
0.8
0.6
0.4
0.2
0.0

```
In [ ]:
```